MIL-S-17243D
21 April 1981
SUPERSEDING
MIL-S-17243C
7 January 1964

MILITARY SPECIFICATION

STONE, COMMUTATOR SURFACING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

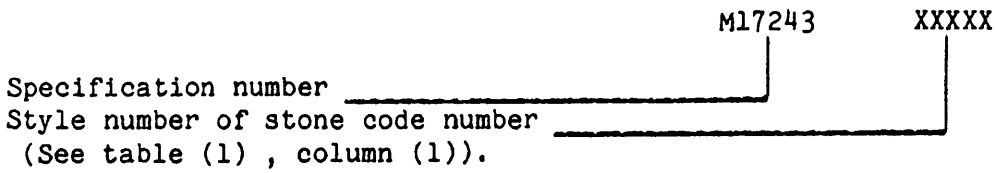
1.1 Scope. This specification covers commutator surfacing stones of the hand grip, unhandled, and the tool post types.

1.2 Classification. Commutator stones shall be furnished in the following classes as specified (see 6.1):

Class EK - Extra Coarse, Very Fast Cutting.
Class K - Coarse, Fast Cutting.
Class M - Medium Cutting.
Class F - Fine, Finishing.
Class EF - Extra-Fine Finishing.

* 1.3 Definitive part number. The definitive part number for items described in this specification was developed as outlined below and defines the requirements of the options presented under this specification (see 6.2). This specification number and the sytle number of stones (column (1) of table (1)) are combined to form the definitive part number.

* 1.3.1 Cataloging data. For cataloging purposes, definitive part numbers for the stones are assigned as follows:



* 1.3.1.1 Style code number. The style code numbers of stones are identified by a five place alpha numeric identifier from column (1) of table I (see example in 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5345

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2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-A-3816 - Abrasives and Abrasive Products; Packaging of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U. S. Property.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC., (ANSI)

B74.12 1974 - Specifications for the Size of ABRASIVE GRAIN - Grinding Wheels, Polishing and General Industrial Use.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

* 3.1 General requirements. Commutator stones specified herein are hand held for the maintenance of commutators and slip-rings on electric motors and generators by grinding, resurfacing and polishing.

* 3.2 Design. The general characteristics, dimensions, and abrasive class shall be as indicated in table I. The abrasive and bonding material shall be electrically non-conductive. Stones shall not drag copper from bar to bar on commutators, nor shall surface short circuits occur at the work face of the stones during resurfacing, grinding, or polishing while the machine is in operation at normal voltage and speed of rotation (see 4.6.1 thru 4.6.4).

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* 3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

* 3.3.1 Abrasive material. The abrasive material shall be aluminum oxide grit. Other abrasive materials may be permitted upon approval of the contracting officer provided the complete commutator stones meet all other requirements of this specification.

* 3.3.2 Bonding material. The abrasive grit shall be bonded together with shellac or other compatible material provided the complete commutator stones meet all other requirements of this specification and no detrimental film or deposit shall remain on the commutator surface or slots.

3.4 Abrasive class. The abrasive size shall be in accordance with ANSI B74.12 by grit size and designated by class as follows:

Class	Size
Extra course (EK)	#24
Course (K)	#46
Medium (M)	#90
Fine (F)	#120
Extra Fine (EF)	#220

Classes EK, K, and M shall be designed to retain sharp cleavage edges in the abrasive grain. Stones shall not drag copper from bar to bar on commutators.

3.4.1 Class EK. Class EK stones shall be suitable for use as a very fast-cutting material where excessive copper must be removed.

3.4.2 Class K. Class K stones shall be suitable for use as a fast-cutting material where excessive copper must be removed.

3.4.3 Class M. Class M stones shall be suitable for use where the mica is slightly high or where the commutator or copper alloy slip ring is slightly rough or grooved and only a light cut need be made.

3.4.4 Class F. Class F stones shall be suitable for use as a fine finishing stone for imparting the final polished surface to a commutator or copper alloy slip ring.

3.4.5 Class EF. Class EF stones shall be suitable for use as an extra-fine finishing stone where it is essential that commutator operate with a minimum of audio and radio noise.

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3.4.6 Classes M/F and F/EF. Stones, combination, classes M/F and F/EF, shall be for varied uses such as where the mica is slightly high, or for imparting the final polished surface to a commutator, or copper alloy slip ring, or for extra fine finishing where it is essential that commutators operate with a minimum of audio noise.

3.5 Sizes and styles. Commutator stones shall be furnished in the style and sizes shown in table I, as specified (see 6.1).

TABLE I. Sizes and styles of stones.

Column 1	Column 2	Column 3	Column 4		
Style number of stones	Figure number illustrating style	Class of abrasive material	Dimensions of stones 1/		
			A	B	C
			Inches	Inches	Inches
00EK1	3	EK	1	1	2
00EK2	4	EK	3	2	3
00EK3	5	EK	5	3	3
00EK4	6	EK	9	5	4
00EK5	8	EK	1-1/2	2-1/2	8
00EK6	8	EK	2	2	8
000K1	3	K	1	1	2
000K2	4	K	2	1-1/2	3
000K3	3	K	3	1	2
000K4	4	K	3	2	3
000K5	4	K	4	3	2
000K6	5	K	5	3	3
000K7	5	K	6	2	3
000K8	5	K	7	3	4
000K9	6	K	9	5	4
00K10	8	K	1	1-1/2	5
00K11	3	K	2	1	2
00K12	8	K	2	2	5
00K13	8	K	1	1-1/2	5
00K14	8	K	2	2	8
00K15	1	K	5/8	3/8	6
00K16	5	K	4	4	3
00K17	5	K	6	4	3
000M1	5	M	4	4	3
000M2	3	M	1	1	2
000M3	4	M	2	1-1/2	2
000M4	4	M	2	2	3
000M5	3	M	3	1	3
000M6	4	M	3	2	2
000M7	4	M	4	3	2
000M8	5	M	5	3	3
000M9	5	M	6	2	3
00M10	5	M	7	3	4
00M11	6	M	9	5	4
00M12	8	M	1	2	8
00M13	5	M	6	4	3
00M14	8	M	2	2	5
00M15	1	M	3/8	5/8	6

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TABLE I. Sizes and styles of stones. (Cont'd)

Column 1	Column 2	Column 3	Column 4		
Style number of stones	Figure number illustrating style	Class of abrasive material	Dimensions of stones 1/ A	B	C
			Inches	Inches	Inches
00M16	3	M	2	1	2
00M17	4	M	2	1-1/2	2
00M18	6	M	10	4	4
00M19	8	M	1	1-1/2	5
00M20	8	M	2	2	8
000F1	5	F	4	4	3
000F2	3	F	1	1	2
000F3	5	F	6	4	3
000F4	4	F	2	2	3
000F5	3	F	3	1	2
000F6	4	F	3	2	3
000F7	4	F	4	2	3
000F8	5	F	5	3	3
000F9	5	F	6	2	3
00F10	5	F	7	3	4
00F11	6	F	9	5	4
00F12	8	F	1	1-1/2	5
00F14	8	F	2	2	5
00F15	1	F	5/8	3/8	6
00F16	3	F	2	1	2
00F17	4	F	2	1-1/2	2
00F18	8	F	1	1-1/2	5
00F19	8	F	2	2	8
00EF1	3	EF	3/4	3/4	2
00EF2	4	EF	2	1-1/2	3
00EF3	3	EF	3	1	3
00EF4	4	EF	3	2	3
00EF5	5	EF	5	3	3
00EF6	8	EF	2	2	5
00EF7	1	EF	5/8	3/8	6
00EF8	2	EF	1	2	8
00EF9	3	EF	1	1	2
0EF10	4	EF	2	2	2
0EF11	6	EF	10	4	4
0EF12	8	EF	1	1-1/2	5
0EF13	8	EF	2	2	8
0M/F2	9	MF	3/4	1/2	3/8
F/EF1	7	F/EF	7/8	5/16	11/16

1/ Refer to figures 1 to 9 inclusive, for features to which dimension letters apply.

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3.6 Standard commercial product. The stone of the same classification shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which, has been sold or is being currently offered for sale, on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.7 Cutting efficiency. Cutting efficiency shall be determined either by weight or by chemical analysis and both before immersion and after immersion in a 20 percent solution of sodium chloride for a period of 24 hours. The efficiency after immersion shall be determined after the stones have dried throughly (see 4.6.2). (Cutting efficiency is the weight of copper particles multiplied by 100, all divided by the weight of the copper particles and disintegrated abrasive produced by the stoning operation.) Under any of the above conditions, the efficiency shall be not less than the following:

Class K - 70 percent.
Class EK - 75 percent.
Class M - 60 percent.
Class F - 60 percent.
Class EF - 55 percent.

3.8 Handles. The unhandled combination and pencil stones shall be similar to figures 1 and 2. All handles used on commutator surfacing stones shall be nonmetallic. For stones to be applied by hand, the handles shall be similar to those shown on figures 3, 4, 5, 6, 7, and 9. Removable round handles for the type commutator surfacing stone shown in figure 4 need not have screw threads provided the design and tolerances are such as to permit secure attachment and prevent the stone from coming loose and falling into the machine. For stones which are to be used in conjunction with mounted resurfacers (precision grinders), the handles shall be similar to that shown on figure 8.

3.8.1 Attachment of handles. Handles shall be attached to stones by means of a sheet or block of electrically nonconducting material securely bonded to the abrasive material. The bonding of the insulation to the abrasive block shall be such that age, moisture, and salt water cannot serve to loosen it. The handle shall be attached to the insulating material in such manner that age, moisture, and salt water cannot loosen it, and that the operator cannot remove it without breaking the handle or the stone (see 4.6.4).

3.9 Identification of product. Unless otherwise specified in contract or order, each commutator stone shall be clearly marked with the style number, and the manufacturer's name and part number in accordance with MIL-STD-130.

* 3.10 Workmanship. Workmanship shall be of the highest professional quality, in all respects, prevailing in the current state-of-the-art.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the

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contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.2.1).
(b) Preparation for delivery inspection (see 4.7).

* 4.2.1 Quality conformance inspection. Quality conformance inspection shall be performed on the stones selected in accordance with 4.4. This inspection shall include the examination of 4.5 and the tests of 4.6.

* 4.3 Inspection lot. All units of the same class, offered to the Government at one time, shall be considered a lot for purposes of inspection. The sample unit shall be one complete commutator stone with handle(s) when applicable.

* 4.4 Sampling. A random sample of stones shall be selected from each lot in accordance with MIL-STD-105.

* 4.4.1 Sampling for examination. Examination of the stones shall be based on inspection level II, and an Acceptable Quality Level (AQL), of 2.5 percent defective.

* 4.4.2 Sampling for test. Tests of the stones shall be based on inspection level S-3 and an AQL of 4.0 percent defective.

* 4.5 Examination. Each sample selected in accordance with 4.4.1 shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. Defects shall be classified in accordance with table II.

TABLE II. Classification of defects in accordance with MIL-STD-105.

Categories	Defects
Critical:	
1	Handle, when required, not a suitable nonmetallic type.
Major:	
101	Class not as specified.
102	Style not as specified.
103	Length not as specified.
104	Width not as specified.
105	Thickness not as specified.
106	Stone chipped, cracked, or damaged.
107	Handle (when requried) separates from the stone without breaking the handle or the stone.

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TABLE II. Classification of defects in accordance with MIL-STD-105. (Cont'd)

Categories	Defects
Minor:	
201	Marking, style number, manufacturer's name and part number missing, incorrect or illegible.

4.6 Tests. Each sample selected in accordance with 4.4.2 shall be tested to determine compliance with this specification. Tests shall be conducted as specified in 4.6.1 through 4.6.4. In lieu of these tests, a test report of previous tests will be acceptable provided the following three conditions are met:

(a) The tests were in accordance with 4.6 for commutator surfacing stones conforming to this specification and the tests were made on a stone of the same class as offered for delivery.
(b) The tests were conducted not more than 5 years prior to date of the applicable contract order.
(c) The tests were witnessed by a Government inspector.

4.6.1 Test procedures. The tests shall be made by either method A or B (see 4.6.2.1 and 4.6.2.2). The dust produced by the stoning operation shall be collected. For classes EK, K and M, following the conditioning operation, the face of the stone shall be examined to determine whether it has picked up copper and to determine whether the stone has worn down in such a manner as to leave continuous sharp cleavage edges in the abrasive grain. The dust shall be examined to determine whether the copper content is in the form of fine dust particles. Where a commutator has been reconditioned, the surface shall be carefully examined to determine whether copper has dragged from bar to bar (see 3.2).

4.6.2 Cutting efficiency. The cutting efficiency of the commutator surfacing stone shall be determined by either of the following two methods as preferred by the contractor; however, the pretesting procedure outlined in 3.7 shall be observed.

4.6.2.1 Method A. The test shall be conducted using a direct current motor or generator having a commutator operating at a peripheral speed of from 720 to 760 feet per minute (fpm). The commutator surfacing stone shall be held against the commutator at sufficient pressure to cut uniformly. The dust shall be collected and the efficiency determined by chemical analysis of a sample of the dust, using nitric acid, 6 normal or stronger.

4.6.2.2 Method B. The test shall be conducted using a cylinder of copper mounted in a lathe so as to simulate a commutator or collector ring. The diameter of the cylinder and the rotational speed of the lathe shall be such as to provide a peripheral speed of from 720 to 760 fpm. The length of the cylinder shall be as required by the width of the stone. The cylinder shall be weighed both before and after grinding to determine the loss in weight. The commutator stone shall also be weighed before and after the grinding to determine the loss in weight. The cutting efficiency may be calculated from these losses in weight, or may be determined by a chemical analysis of a sample of the dust, using nitric acid, 6 normal or stronger.

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4.6.3 Conductivity test (mounted stones only). The stones shall be immersed (see 3.7) in salt water and then dried thoroughly. After the drying period, tests shall be conducted to determine whether the material which bonds the abrasive to the handle is nonconducting.

4.6.4 Drop test for stones with handles. Stones with handles, of each representative size shall be dropped in three different positions through a distance of 3 feet to a steel floor to determine whether the handles of mounted stones are adequately secured. The handle must not loosen from the stone nor shall it be possible to remove the handle without breaking the stone or the handle (see 3.8.1).

* 4.7 Packaging inspection. The preservation, packing, and marking of the stones shall be inspected to verify conformance to the requirements of section 5.

* 5. PACKAGING

* 5.1 Preservation, packing and marking. The commutator stones shall be preserved, packed and marked in accordance with MIL-A-3816. The level of preservation and level of packing shall be as specified (see 6.1).

6. NOTES

* 6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2 and table I).
(c) Size and style required (see 3.5).
(d) Level of preservation and level of packing required (see 5.1).

* 6.2 Example of definitive part number.

M17243 - 00EK1

This specification number ——————
Five place alpha numeric identifier ——————
taken from column 1 of table I
(see 1.2, 1.3, 1.3.1, 1.3.1.1, and 3.5).

* 6.3 Significant changes. Significant changes to the previous issue are indicated by an asterisk near the paragraph number.

6.4 Illustrations. The illustrations herein are descriptive and not restrictive, and are not intended to preclude the purchase of commutator resurfacing stones otherwise conforming to this specification.

Custodians:

Army - AR
Navy - YD
Air Force - 99

Preparing activity:

Navy - YD

Project No. 5345-0034

User activities:

Army - ER, ME

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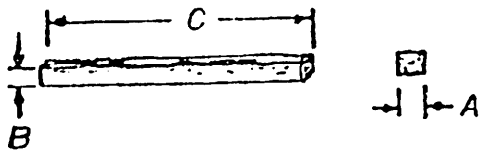


Figure 1 Pencil style, commutator surfacing stone.

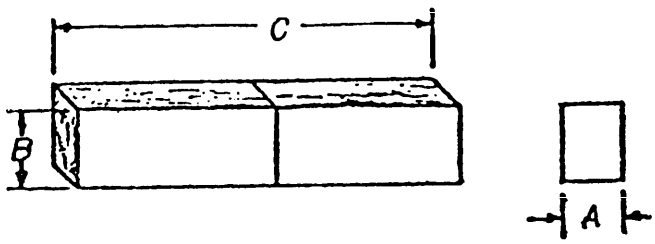


Figure 2 Pocket style, combination commutator surfacing stone.

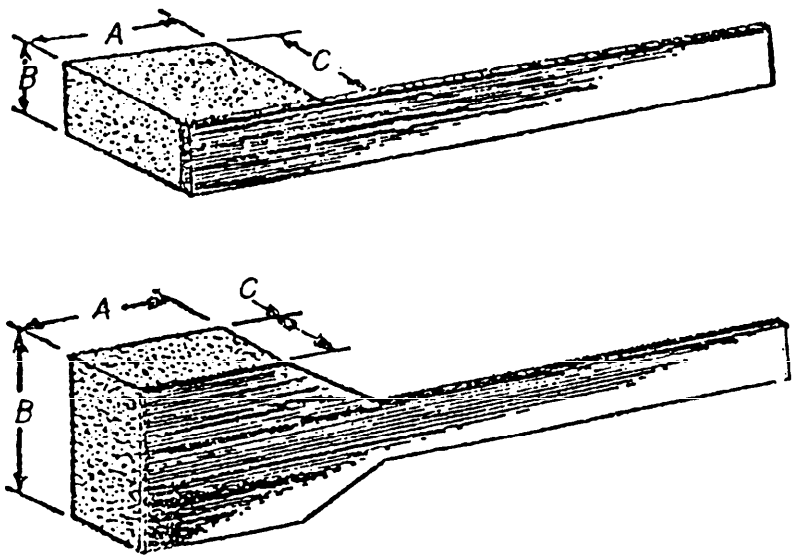


Figure 3 Straight handle commutator surfacing stone.

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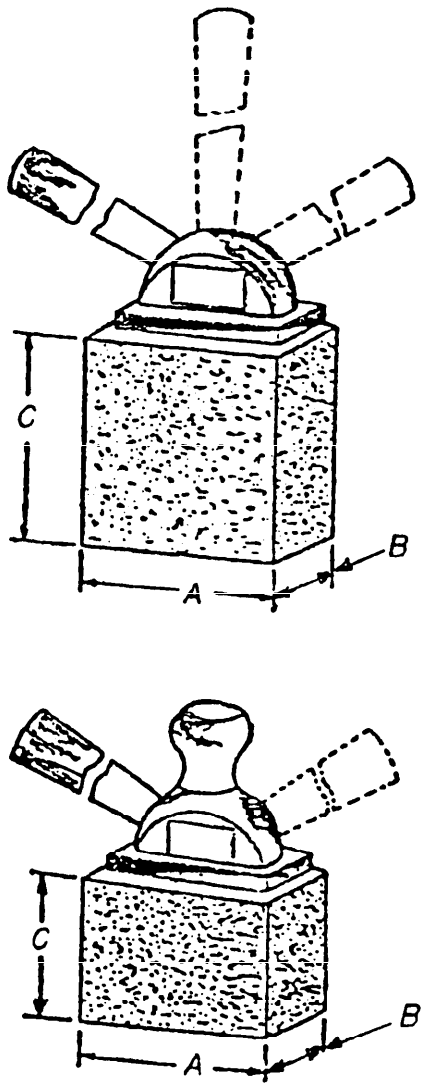


Figure 4 Adjustable handle commutator surfacing stone.

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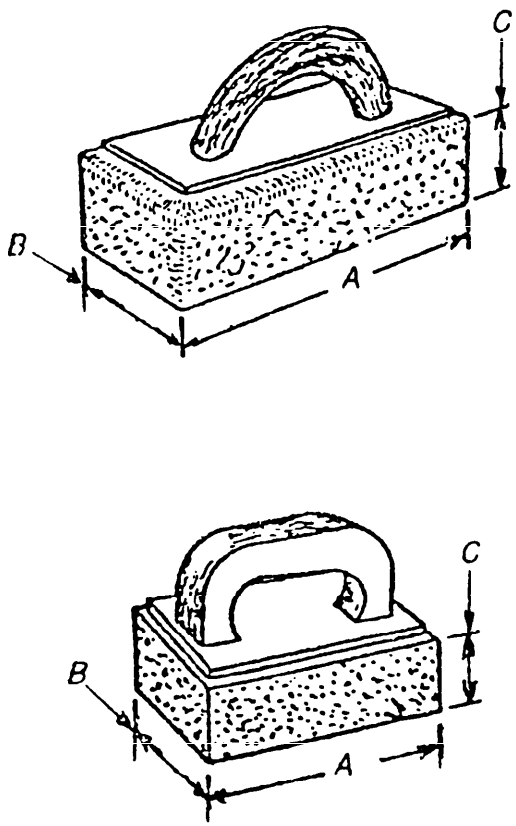


Figure 5 Single-grip handle commutator surfacing stone.

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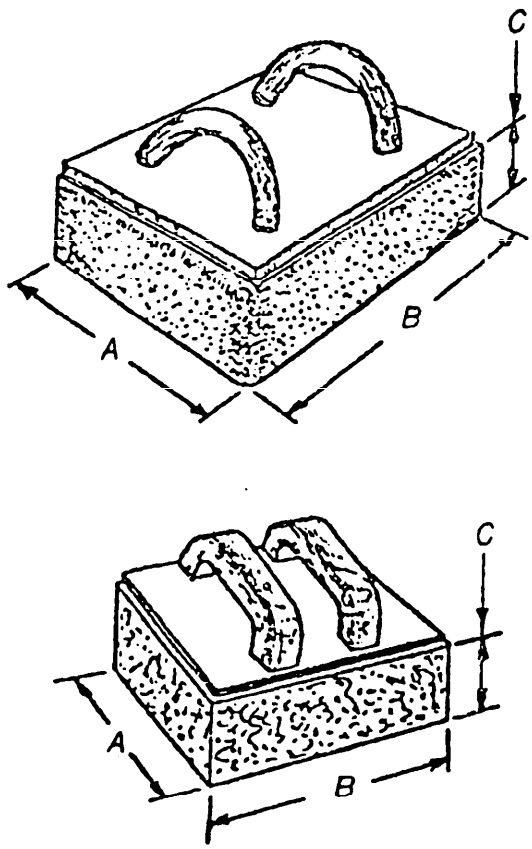


Figure 6 Double-grip handle commutator surfacing stone.

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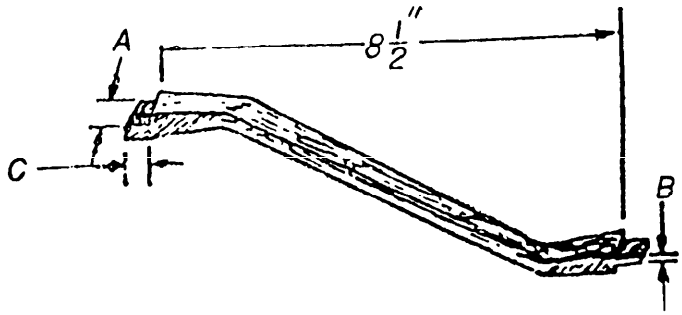


Figure 7 Handled style, combination commutator surfacing stone.

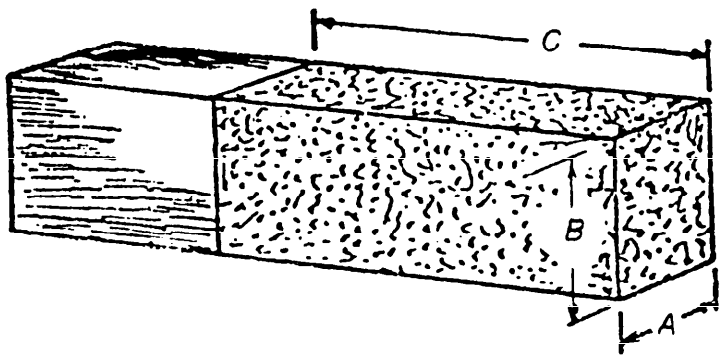


Figure 8 Tool post handle commutator surfacing stone.

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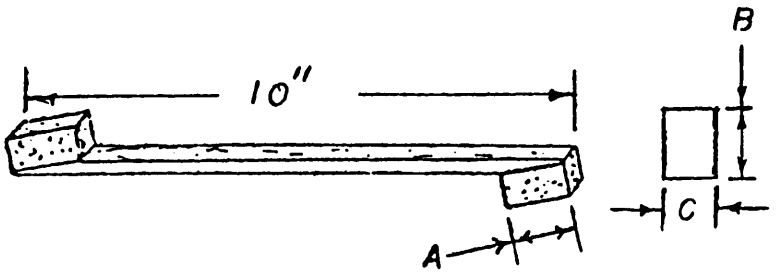


Figure 9 Handled style, small motor type commutator surfacing stone.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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